

Osmar Alejandro Liera Gómez

Applied AI Engineer · LLM Agents & Production AI Integration

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PROFESSIONAL SUMMARY

Software engineer focused on **putting AI to work inside real business systems**. I design and ship **LLM agents with tool calling, structured outputs and human-in-the-loop approval**, and integrate them with enterprise data and APIs. At ELROI Labs I built the AI layer of our CRM: an **MCP server exposing 23 tools** that lets AI assistants query the pipeline, plan follow-ups and act on records under the user's own permissions, with every action audited. At Nuts Marketing I build Shopify ↔ Microsoft Dynamics 365 middleware. At Tangramx I work in **Python (FastAPI)**, where I shipped **LLM-based OCR to production** and **evaluated it against ground truth** before rollout. At the Instituto Tecnológico de La Paz I program institutional systems and teach software development.

TECHNICAL SKILLS

Agents	Tool calling, MCP servers (tools, prompts, resources), human-in-the-loop approval, conversation state	LLMs	OpenAI (Vision, Responses API), structured outputs via JSON Schema, prompt design & guardrails
Evaluation	Ground-truth validation, deterministic post-processing of model output, data-quality analysis	Python	FastAPI, SQLAlchemy, Alembic, REST APIs, JWT + 2FA
Integration	REST APIs, OpenAI-compatible model gateways, WhatsApp messaging API, Shopify Admin API, Magento, Dynamics 365, SOAP	Data	SQL, MySQL, SQLite, data modeling, versioned migrations, Google Document AI
Also	Laravel (PHP), Nova, Filament, React 18, TypeScript, Tailwind CSS, automated feature testing	Cloud / Ops	Azure (App Service, WebJobs), Docker, Linux, Git/GitHub, CI/CD
Languages	Spanish (native) · English intermediate (B1–B2): daily technical reading and written communication		

PROFESSIONAL EXPERIENCE

Full-Stack Developer 2024 — Present
Nuts Marketing

Integration middleware between Shopify stores and Microsoft Dynamics 365 ERPs, plus the Shasa (Mexico and US) and Hypsters platforms. Stack: Laravel, Laravel Nova, Filament, Shopify Admin API (GraphQL and webhooks), SOAP and Azure App Service with WebJobs.

- **Macadamia (internal project):** Shopify ↔ Dynamics 365 Finance & Operations middleware for a retailer: orders, customers, store pickup by location, payment mapping and Shopify ↔ middleware ↔ F&O reconciliation reports.
- **El Asturiano:** Dynamics 365 Business Central + LS Central over SOAP/XML; receives Shopify webhooks and syncs inventory with scheduled jobs.
- **Shasa (Mexico and US) and Hypsters:** Shopify and Magento integrations for catalogs, inventory, orders and customers; Apple Wallet passes and gift cards. **Coco (internal project):** new Laravel + Nova app on Azure App Service.

Founder & Technical Lead Present
ELROI Labs — La Paz, BCS

Software, automation and applied-AI studio. I lead technical operations and deliver projects end to end.

- OS CRM — AI agent layer**
Internal CRM for leads, pipeline, multichannel follow-up, clients and projects (Laravel, React)
- Built an **MCP server with 23 tools, 3 prompts and a playbook resource** so AI assistants (Claude and others) can search leads, plan the day, log follow-ups, schedule tasks, import leads and generate pipeline and activity reports.
 - Agents run with **the token owner's permissions**, every change is written to the audit log, and outbound WhatsApp messages require **explicit human approval** before sending.
 - Developed **LLM agents with structured outputs** (JSON Schema): a lead-contact assistant that drafts messages, call scripts and next tasks; a project planner that breaks scope into tasks with acceptance criteria and estimates; and a WhatsApp writer grounded in the real conversation and an editable business context.
 - Designed **guardrails against hallucination**: answer only from supplied data, ask instead of assuming, no prices or commitments, per-agent temperature and token limits.
 - Wrote a **custom OpenAI-compatible model gateway** with model-catalog discovery, caching and fallback; multi-provider setup (OpenAI, Anthropic, Gemini, Azure and others).
 - Covered the MCP server, AI assistant and WhatsApp flows with **automated feature tests**.
- Client work**
- **Applied AI** for customer service, lead prequalification and follow-up automation.

Tangramx

Management applications for insurers, healthcare institutions and logistics operators. Primary stack: Python / FastAPI.

Vitase / Paciente Integral — LLM-based document OCR

Medical records and claims management for insurers (Jalisco Health Services)

- Integrated **OpenAI Vision** for OCR of Mexican national ID cards (INE), replacing the previous engine.
- Built a **deterministic validation layer on top of the model output** for CURP: check-digit reconstruction, homoclave recovery and edge cases (Ñ, MARÍA/JOSÉ compound names).
- **Evaluated against ground truth**: 25 real ID cards with **100% name accuracy**; documented the data-quality findings that guided rollout.

ATC — Account Statement Processor

Automated processing of medical account statements (FastAPI)

- Fixed incorrect discount detection and an off-by-one date shift in the parser; wrote the local setup guides.

Compass WMS · ATCSMART

Warehouse management and insurance-claims systems

- Implemented **true server-side pagination**, fixed a **race condition** in multi-file upload, and added Excel exports segmented by insurer and hospital.

SYSTEMS & TEACHING

Institutional Systems Programmer & Teacher

Present

Instituto Tecnológico de La Paz

- **Programmer** of the Tec’s institutional systems.
- **Teacher** in the final semester of the software development programs; as part of teaching, training on applied AI for business.

PREVIOUS EXPERIENCE

- **Giro26** — Full-Stack Developer: financial ERP, hexagonal architecture and SQL optimization.
- **Hi-G** — Web & Mobile Developer: Hi-G app with Flutter, Firebase, TypeScript and Web3 integrations.

PERSONAL PRODUCTS

PMCRM

Workspace for managing projects, files, notes, clients and collaboration from a single platform.

AVT

My own multitenant SaaS for land sales: users, plots, zones, maps and commercial operations.

NopalGreen

Operations platform for tortilla producers: POS, production, inventory, delivery and reporting.